SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : AI

Aim: Naive Bayes' Classifier
Implement the Naive Bayes' algorithm for classification.
Train a Naive Bayes' model using a given dataset and calculate class probabilities.
Evaluate the accuracy of the model on test data and analyze the results.

Input

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```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report

# Load dataset
df = pd.read_csv("confidence_features.csv")

print("Dataset shape:", df.shape)

print("\nColumns:")
print(df.columns)

# Remove missing values
df = df.dropna()

# Target column
target = "confidence_label"

# Separate input and target
X = df.drop(columns=[target])
y = df[target]

# Convert categorical features into numbers
categorical_columns = X.select_dtypes(include=["object"]).columns

for column in categorical_columns:
    encoder = LabelEncoder()
    X[column] = encoder.fit_transform(X[column])

# Convert target labels into numbers
target_encoder = LabelEncoder()
y = target_encoder.fit_transform(y)
```

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```python
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining records:", len(X_train))
print("Testing records:", len(X_test))

# Create Gaussian Naive Bayes model
model = GaussianNB()

# Train model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nNaive Bayes Accuracy:",
      round(accuracy * 100, 2), "%")

# Calculate class probabilities
probabilities = model.predict_proba(X_test)

# Display probabilities for first 5 test records
print("\nClass Probabilities:")

for i in range(5):

    print("\nTest Record", i + 1)
```

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```
    for j in range(len(target_encoder.classes_)):
        print(
            target_encoder.classes_[j],
            ":",
            round(probabilities[i][j] * 100, 2),
            "%"
        )

# Classification report
print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=target_encoder.classes_,
        zero_division=0
    )
)

# Probability graph for first test record
first_probabilities = probabilities[0] * 100
class_names = target_encoder.classes_

plt.figure(figsize=(7, 5))

bars = plt.bar(
    class_names,
    first_probabilities
)

plt.title("Naive Bayes Class Probabilities")
plt.xlabel("Confidence Class")
plt.ylabel("Probability (%)")

plt.ylim(0, 100)
```

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```
print(
    classification_report(
        y_test,
        y_pred,
        target_names=target_encoder.classes_,
        zero_division=0
    )
)

# Probability graph for first test record
first_probabilities = probabilities[0] * 100
class_names = target_encoder.classes_

plt.figure(figsize=(7, 5))

bars = plt.bar(
    class_names,
    first_probabilities
)

plt.title("Naive Bayes Class Probabilities")
plt.xlabel("Confidence Class")
plt.ylabel("Probability (%)")

plt.ylim(0, 100)

# Display probability values
for bar, value in zip(bars, first_probabilities):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        value + 2,
        f"{value:.2f}%",
        ha="center"
    )

plt.show()
```

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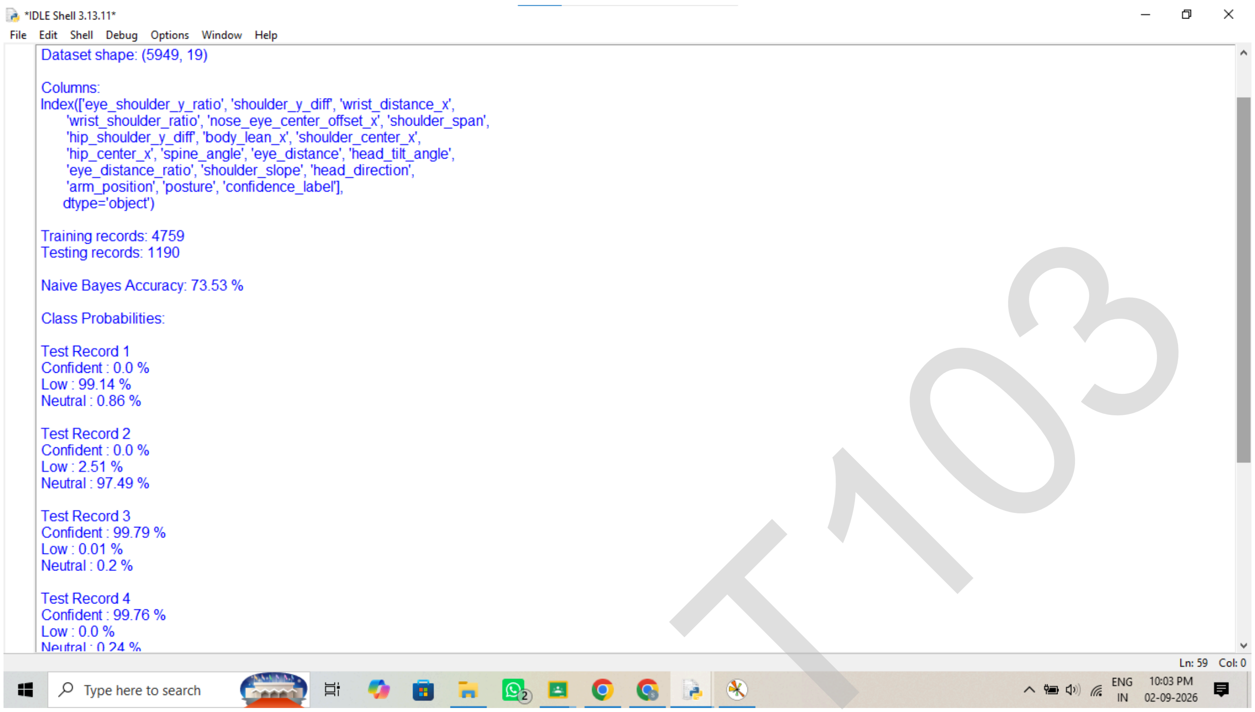
Output



```
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Dataset shape: (5949, 19)

Columns:
Index(['eye_shoulder_y_ratio', 'shoulder_y_diff', 'wrist_distance_x',
       'wrist_shoulder_ratio', 'nose_eye_center_offset_x', 'shoulder_span',
       'hip_shoulder_y_diff', 'body_lean_x', 'shoulder_center_x',
       'hip_center_x', 'spine_angle', 'eye_distance', 'head_tilt_angle',
       'eye_distance_ratio', 'shoulder_slope', 'head_direction',
       'arm_position', 'posture', 'confidence_label'],
      dtype='object')

Training records: 4759
Testing records: 1190

Naive Bayes Accuracy: 73.53 %

Class Probabilities:

Test Record 1
Confident : 0.0 %
Low : 99.14 %
Neutral : 0.86 %

Test Record 2
Confident : 0.0 %
Low : 2.51 %
Neutral : 97.49 %

Test Record 3
Confident : 99.79 %
Low : 0.01 %
Neutral : 0.2 %

Test Record 4
Confident : 99.76 %
Low : 0.0 %
Neutral : 0.24 %

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```
Test Record 4
Confident : 99.76 %
Low : 0.0 %
Neutral : 0.24 %

Test Record 5
Confident : 99.76 %
Low : 0.05 %
Neutral : 0.19 %

Classification Report:
              precision    recall  f1-score   support

   Confident       0.79      0.86      0.83       627
         Low       0.75      0.81      0.78       231
     Neutral       0.57      0.44      0.50       332

    accuracy                           0.74      1190
   macro avg       0.70      0.71      0.70      1190
weighted avg       0.72      0.74      0.73      1190

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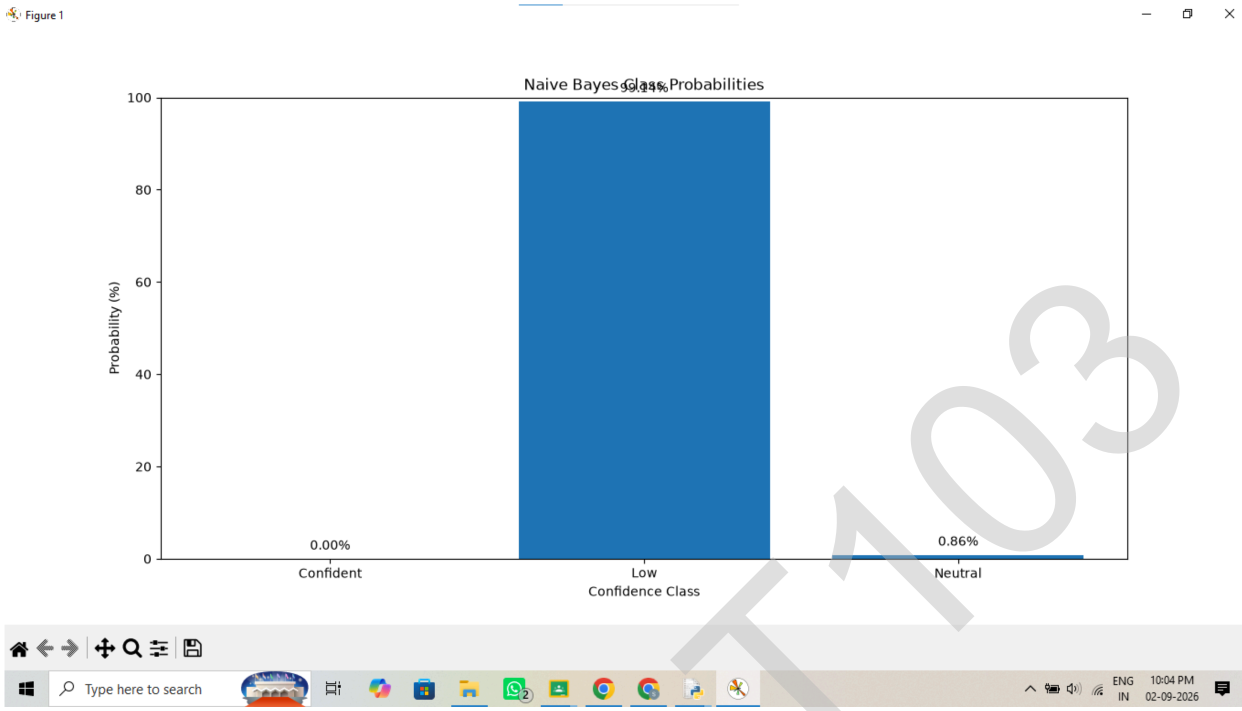

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